UNITED STATES
NUCLEAR REGULATORY COMMISSION
REGION III
2056 WESTINGS AVENUE, SUITE 400
NAPERVILLE, IL 60563-2657

May 21, 2025

Mike Mlynarek
Site Vice President
Holtec Decommissioning
International, LLC
Palisades Nuclear Plant
27780 Blue Star Memorial Highway
Covert, MI 49043-9530

SUBJECT: PALISADES NUCLEAR PLANT – RESTART INSPECTION REPORT 05000255/2025002

Dear Mike Mlynarek:

On April 30, 2025, the U.S. Nuclear Regulatory Commission (NRC) completed an inspection at the Palisades Nuclear Plant. The enclosed inspection report documents the inspection results, which the inspectors discussed on May 13, 2025, with yourself and other members of your staff.

No violations of more than minor significance were identified during this inspection.

On June 13, 2022, Palisades ceased permanent power operations and subsequently removed all fuel from the reactor, as detailed in the letter from Entergy to the NRC, "Certifications of Permanent Cessation of Power Operations and Permanent Removal of Fuel from the Reactor Vessel," (ADAMS Accession No. ML22164A067). On September 28, 2023, Holtec Decommissioning International, LLC (Holtec) submitted a letter to the NRC requesting exemptions from certain portions of the requirements of Title 10 of the *Code of Federal Regulations* (10 CFR) 50.82(a)(2) to pursue the potential reauthorization of power operations at Palisades (ADAMS Accession No. ML23271A140). The NRC's initial acceptance to review Palisades' request for exemptions was documented in ADAMS at Accession No. ML23291A440 on November 3, 2023.

This letter, its enclosure, and your response (if any) will be made available for public inspection and copying at http://www.nrc.gov/reading-rm/adams.html and at the NRC Public Document Room in accordance with Title 10 of the *Code of Federal Regulations* 2.390, “Public Inspections, Exemptions, Requests for Withholding.”

Sincerely,

April M. Nguyen Signed by Nguyen, April on 05/21/25

April M. Nguyen, Lead
Palisades Restart Team
Division of Operating Reactor Safety

Docket No. 05000255
License No. DPR–20

Enclosure:
As stated

cc w/ encl: Distribution via LISTSERV®

Letter to Mike Mlynarek from April Nguyen dated May 21, 2025.

SUBJECT: PALISADES NUCLEAR PLANT – RESTART INSPECTION REPORT 05000255/2025002

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U.S. NUCLEAR REGULATORY COMMISSION
Inspection Report

Docket Numbers:	05000255
License Numbers:	DPR-20
Report Numbers:	05000255/2025002
Enterprise Identifiers:	I-2025-002-0066
Licensee:	Holtec Decommissioning International, LLC
Facility:	Palisades Nuclear Plant
Location:	Covert, MI
Inspection Dates:	March 01, 2025, to April 30, 2025
Inspectors:	B. Bergeon, Senior Operations Engineer K. Carrington, Senior Resident Inspector, North Anna N. Hansing, Mechanical Engineer, NRR D. Hoang, Senior Project Engineer D. Lanyi, Senior Operations Engineer K. Kirchbaum, Senior Operations Engineer J. Mancuso, Senior Resident Inspector P. Meier, Senior Resident Inspector, Sequoyah V. Myers, Senior Health Physicist T. Okamoto, Resident Inspector J. Robb, Operations Engineer C. St. Peters, Senior Project Engineer A. Shaikh, Senior Reactor Inspector C. Speer, Reactor Systems Engineer, NRR J. Winslow, Senior Project Engineer, Restart Team
Approved By:	April M. Nguyen, Lead Palisades Restart Team Division of Operating Reactor Safety

Enclosure

SUMMARY

The U.S. Nuclear Regulatory Commission (NRC) performed inspections of the licensee's activities to verify that the reactor coolant system boundary, reactor vessel head, and risk-significant piping system boundaries are appropriately monitored for degradation and that required repairs and replacements are appropriately examined and accepted; systems, structures, and components (SSCs) are appropriately inspected, maintained, and tested to ensure they can perform their required safety functions for potential restart; the licensed operator requalification training program and operator performance are appropriately performed; and radiation safety activities are appropriately performed.

List of Findings and Violations

None.

Additional Tracking Items

None.

PLANT STATUS

The Palisades Nuclear Plant ceased power operations and subsequently removed all fuel from the reactor, as certified in a letter to the NRC on June 13, 2022. On September 28, 2023, Holtec Decommissioning International, LLC (Holtec) submitted a letter to the NRC requesting exemptions from certain portions of the requirements of 10 CFR 50.82(a)(2) to pursue the potential reauthorization of power operations. The NRC is conducting inspections to evaluate and assess the licensee's activities for potential restoration to an operational status.

The resident inspectors, as well as supplemental inspectors, continue to supply daily oversight of the restart activities occurring at the site.

INSPECTION SCOPES

Inspections were conducted using the appropriate portions of the referenced inspection procedures (IPs) in effect at the beginning of the inspection unless otherwise noted. Currently approved IPs with their attached revision histories are located on the public website at http://www.nrc.gov/reading-rm/doc-collections/insp-manual/inspection-procedure/index.html.

REACTOR SAFETY

71111.08P - Inservice Inspection Activities (PWR)

From April 28, 2025, to May 1, 2025, the inspectors observed and verified Holtec Palisades' nondestructive examination (NDE) and repair/replacement activities of the reactor vessel closure head, Alloy 600 weld mitigation activities, and steam generator in-situ pressure testing, conducted in accordance with the requirements of the ASME Code Section XI. The purpose of these inspections was to verify that the reactor coolant system boundary, reactor vessel closure head, steam generators, and risk-significant piping system boundaries are appropriately monitored for degradation. The inspectors also verified that the examination results were thoroughly reviewed and appropriately evaluated, and that required corrective actions to address identified indications and defects will be appropriately implemented. During this quarter, the licensee completed NDE activities and evaluation of data for the reactor vessel closure head, steam generator tubes, and Alloy 600 welds in the primary coolant system and continued repair/replacement activities in all areas. Further NRC inspections will monitor repair/replacement activities as they are conducted.

The inspectors verified that the following NDE and repair/replacement activities were conducted appropriately per the ASME Code or required standard, and that any potential indications and defects were identified and evaluated at the proper thresholds:

Nondestructive Examination and Welding Activities (IP Section 03.01)

- Safety Injection Cold Leg 1A&B and 2A&B Alloy 600 mitigation Weld Overlay Fabrication
- Reactor Vessel Internal (RVI) indications analysis and NDE procedures

Vessel Upper Head Penetration Inspection Activities (IP Section 03.02)

- Alloy 690 reactor head repair nozzle Mill Report and receipt inspection data

Steam Generator Tube Inspection Activities (IP Section 03.04)

- Steam generator A in-situ pressure test
- Steam generator A test of sleeving repair methodology and Eddy Current examinations

71111.11A - Licensed Operator Requalification Program and Licensed Operator Performance

Licensed Operator Requalification Program and Licensed Operator Performance

(1) The inspectors reviewed and evaluated the licensed operator examination failure rates for the requalification annual operating exam administered from February 25, 2025, to March 28, 2025.

71111.13 – Maintenance Risk Assessments and Emergent Work Control

Risk Assessment and Management Sample (IP Section 03.01)

The inspectors evaluated the accuracy and completeness of risk assessments for the following planned work activities to ensure configuration changes and appropriate work controls were addressed:

(1) Low pressure “A” and “B” main turbine rotor lift activities on April 11, 2025
(2) Primary coolant pump “B” lift activities on April 1, 2025
(3) Steam generator “A” nozzle dam installation and primary coolant system inventory control activities on April 18, 2025

RADIATION SAFETY

Radiation Safety – Occupational

From April 21 to April 25, 2025, the inspectors evaluated various ongoing radiologically risk-significant work evolutions, including steam generator primary side work and reactor head work. For these work evolutions, the inspectors observed activities such as extremity dosimetry issuance and use, respiratory protection issuance and use, use of temporary ventilation systems, high radiation area and radiation work permit briefings, and radiation protection job coverage. The inspectors also walked down the auxiliary building and containment to observe radiological postings and controls. The inspectors also reviewed various dose assessments for previous work activities. This review included quality control procedures and records for use of the whole-body counter, whole body count results, internal dose assessments, evaluation of multiple dosimetry use, and evaluation of effective dose equivalence use and calculations. The inspectors conducted these reviews using IP 71124.01, “Radiological Hazard Assessment and Exposure Controls,” IP 71124.03, “In-Plant Airborne Radioactivity Control and Mitigation,” and IP 71124.04, “Occupational Dose Assessment,” as applicable.

OTHER BASELINE ACTIVITIES

71152 – Problem Identification and Resolution

System Return to Service

During the inspection period, the inspectors continued reviews of Phases I and II of the licensee's system return to service (SRTS) plans. The SRTS plans document licensee activities to verify that the configuration and condition of Systems, Structures, and Components (SSCs) are consistent with the design and licensing bases to support potential restart. The inspectors reviewed Phases I and II of the SRTS plans, which include areas such as licensing and design-basis functions, system material conditions, reviewing open and deferred work prior to shutdown, and program applicability. The inspectors conducted these reviews using IP 71152, "Problem Identification and Resolution," and IP 71111.21M, "Comprehensive Engineering Team Inspection (CETI)," as applicable.

The inspectors commenced reviews of the SRTS plans for the following systems:

- Containment building and miscellaneous equipment
- Containment purge
- Containment spray
- Chemical and volume control
- Circulating water
- High pressure safety injection
- Instrument air and service air
- Main feedwater
- Primary coolant
- Pressurizer pressure and level control
- Safety injection and refueling water tank and containment sump
- Safety injection tanks
- Radiation Monitoring System

The inspectors completed initial reviews of the SRTS plans for the following systems:

- Containment and Fuel Pool Cranes
- Control rod drive
- Control room heating, ventilation, and air conditioning (HVAC)
- Dry fuel storage
- Reactor cavity flood
- Spent fuel pool cooling
- Ultimate heat sink

These systems were selected based on the importance of the SSCs during normal and emergency plant operating conditions, risk significance and defense-in-depth of the SSC's specific functions and required work/modifications of the SSCs scheduled prior to potential restart.

The inspectors evaluated the initial SRTS plans for each system to identify if planned licensee activities will verify that the configuration and condition of the SSCs are consistent with the proposed operational design and licensing bases to support restart and verify readiness of the SSCs to support a return to service. The inspectors noted that the SRTS plans are intended to be living documents with a phased approach to SRTS. The initial plans were determined to be at a very high level and did not always provide specific details associated with inspection or testing plans. The inspectors understood that this specific level of detail is still in progress and, as the licensee progresses through the SRTS phases, the SRTS plans will undergo additional reviews by NRC inspector to ensure appropriate SSC restoration and operability/functionality.

INSPECTION RESULTS

No additional inspection results.

EXIT MEETINGS AND DEBRIEFS

The inspectors verified no proprietary information was retained or documented in this report.

- On May 13, 2025, the inspectors presented the inspection results to M. Mlynarek, Site Vice President, and other members of the licensee staff.
- On April 14, 2025, the inspectors presented the licensed operator requalification program inspection results to C. Sizemore, Operations Training Superintendent, and other members of the licensee staff.

DOCUMENTS REVIEWED

Inspection Procedure	Type	Designation	Description or Title	Revision or Date
71111.08P	Corrective Action Documents Resulting from Inspection	2025-0473	A Minor Correction to NDE Report HLSC-350-00-PT-002 Concerning the Material Type	02/21/2025
	Procedures	03-601600	Operating Instructions for In Situ Pressure Testing	010
		55-OI0033-022	Ambient I.D.T.B Welding of CEDM/CET Penetration of Reactor Vessel Closure Head Utilizing Modified Local Cavity Weld Head	03/19/2025
	Shipping Records	1024018714	Framatome Purchase Order QA Data Package SB-167 UNS N06690 Seamless Tubing	007
		38001288	Holtec Purchase Order QA Data Package for Replacement CRDM Nozzles and Replacement ICI Nozzles	001
71111.20	Procedures	FHS-M-23	Movement of Heavy Loads in the Spent Fuel Pool Area	Revision 42
		FHS-M-24	Movement of Heavy Loads in the Containment Building Area	Revision 42
		MSM-M-72	Movement of Heavy Loads in the Turbine Building	Revision 42
		RFL-SG-2	S/G Primary Nozzle Dam Installation and Removal	Revision 11
	Work Orders	WO# 00586667	E-50A INSTALL/REMOVE NOZZLE DAMS PER RFL-SG-2	04/18/2025
71124.01	ALARA Plans	20250454	ALARA Plan for RWP 20250454	0
		20250454	ALARA Plan for RWP 20250454	0
	Radiation Surveys	2025-0802	“A” Steam Generator Platform Survey	04/19/2025
		2025-0816	“A” Steam Generator Platform Survey	04/21/2025
	Radiation Work Permits (RWPs)	20250450	Reactor Head, Nozzle Machining, Nozzle Replacement, Decon and Associated Tasks	0
		20250454	Steam Generator Primary Side Activities	0
71124.03	Miscellaneous	N/A	Grade D breathing Air Quality Test	04/17/2025
71124.04	Miscellaneous	N/A	Internal Dose Assessment	09/06/2024
		N/A	Internal Dose Assessment	09/08/2024
	Procedures	EN-RP-208	Whole-Body Counting/In-Vitro Bioassay	7
		HP 8.11	Whole-Body Counting	18
71152	Calculations	GWO-6861CALC2	Proposed Repair Program for the Auxiliary Building SIRW Tank Cellular Slab at Palisades Nuclear Power Plant	0

Inspection Procedure	Type	Designation	Description or Title	Revision or Date
	Corrective Action Documents	PAL-05382	Main Control Roof Cellular Slab	02/11/2025
		PAL-05888	Typographical Error in Report to NRC	03/11/2025
	Engineering Evaluations	PLP-RPT-18-0024	1R26 Refueling Outage Safety Related Containment Coating Assessment Report	02/06/2019
	Miscellaneous	PLP-SRTSP-CLC	Containment and Fuel Pool Cranes	Draft B
		PLP-SRTSP-CLP	Containment Building and Miscellaneous Equipment	Draft B
		PLP-SRTSP-CPG	Containment Purge	Draft A
		PLP-SRTSP-CRD	Control Rod Drive	Draft A
		PLP-SRTSP-CRV	Control Room HVAC	Draft A
		PLP-SRTSP-CSS	Containment Spray System	Draft A
		PLP-SRTSP-CVC-CBA	Chemical and Volume Control	Draft A
		PLP-SRTSP-CWS	Circulating Water	Draft B
		PLP-SRTSP-DFS	Dry Fuel Storage	Draft A
		PLP-SRTSP-HPI	High Pressure Safety Injection	Draft A
		PLP-SRTSP-IAS	Instrument Air and Service Air	Draft A
		PLP-SRTSP-MFW	Main Feedwater	Draft A
		PLP-SRTSP-PCS	Primary Coolant	Draft A
		PLP-SRTSP-PZR	Pressurizer Pressure and Level Control	Draft A
		PLP-SRTSP-RCF	Reactor Cavity Flood	Draft A
		PLP-SRTSP-RIA	Radiation Monitoring System Return-To-Service Plan	Draft A
		PLP-SRTSP-SFP	Spent Fuel Pool Cooling	Draft A
		PLP-SRTSP-SIT	Safety Injection Tanks	Draft A
		PLP-SRTSP-SSS	SIRW and Containment Sump system	Draft A
		PLP-SRTSP-UHS	Ultimate Heat Sink	Draft A
		PNP 2025-005	50–Year Containment Tendon Surveillance Inspection Summary Report	03/03/2025
	Procedures	EA-CAP-047706-01/0	Palisades Reactor Head Drop Analysis	02/02/2006
		MSM-M-13A/9	L-1 Periodic Inspection	09/28/2023
		MSM-M-13B/11	L-2 Periodic Inspection	02/05/2024
		MSM-M-13C/11	L-3 Periodic Inspection	05/13/2023

Inspection Procedure	Type	Designation	Description or Title	Revision or Date
		WI-MA-349/0	L-2 Periodic Inspection for Electrical and Mechanical Components	12/18/2024
	Work Orders	WO 50083288-01-001	Containment Polar Crane	10/17/2023